

DSCI 6700 文本挖掘与非结构化数据

作业 2.1 —— 词干提取对比 (Porter / Snowball / Lancaster)

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课程：DSCI 6700 文本挖掘与非结构化数据

作业：2.1 词干提取

这次作业我是用 Python 的 NLTK 库 (nltk.stem) 来跑的，里面正好有课程里讲的 Porter、Snowball 和 Lancaster 三个提取器，默认那个就是 Porter。我先把文本都转成小写、去掉标点，再一个词一个词地提取。提取的是英文单词，所以下面的原文和结果我都保留了英文，只有说明和分析用中文写。

第一部分 里根告别演说英文段落

下面这段文字选自里根总统《告别演说》（1989 年 1 月 11 日）中“常识 (common sense)”那一段，使用英文原版。

1.1 原文 (提取词干前)

Common sense told us that when you put a big tax on something, the people will produce less of it. So, we cut the people's tax rates, and the people produced more than ever before. The economy bloomed like a plant that had been cut back and could now grow quicker and stronger. Our economic program brought about the longest peacetime expansion in our history: real family income up, the poverty rate down, entrepreneurship booming, and an explosion in research and new technology. We're exporting more than ever because American industry became more competitive.

1.2 分词后 (小写) 文本

common sense told us that when you put a big tax on something the people will produce less of it so we cut the people's tax rates and the people produced more than ever before the economy bloomed like a plant that had been cut back and could now grow quicker and stronger our economic program brought about the longest peacetime expansion in our history real family income up the poverty rate down entrepreneurship booming and an explosion in research and new technology we're exporting more than ever because american industry became more competitive

1.3 三种提取器的输出

Porter 提取器：

common sens told us that when you put a big tax on someth the peopl will produc less of it so we cut the people' tax rate and the peopl produc more than ever befor the economi bloom like a plant that had been cut back and could now grow quicker and stronger our econom program brought about the longest peacetim expans our histori real famili incom up the poverti rate down entrepreneurship boom and an explos in research and new technolog we'r export more than ever becaus american industri becam more competit

Snowball 提取器：

common sens told us that when you put a big tax on someth the peopl will produc less of it so we cut the peopl tax rate and the peopl produc more than ever befor the economi bloom like a plant that had been cut back and could now grow quicker and stronger our econom program brought about the longest peacetim expans our histori real famili incom up the poverti rate down entrepreneurship boom and an explos in research and new technolog we'r export more than ever becaus american industri becam more competit

Lancaster 提取器：

common sens told us that when you put a big tax on someth the peopl wil produc less of it so we cut the people's tax rat and the peopl produc mor than ev bef the econom bloom lik a plant that had been cut back could now grow quick and stronger our econom program brought about the longest peacetim expand in our his real famy incom up the poverty rat down entrepr boom and an explod in research and new technolog we'r exp mor than ev becaus am industry becam mor competit

1.4 三种提取器结果不一致的单词

共 94 个词，其中 38 个词被至少一种提取器改变，有 21 个词在三种算法下得到了不同的词干。部分示例：

原词	Porter	Snowball	Lancaster
will	will	will	wil
people's	people'	peopl	people's
rates	rate	rate	rat
more	more	more	mor
ever	ever	ever	ev
before	befor	befor	bef
economy	economi	economi	econom
like	like	like	lik
quicker	quicker	quicker	quick
expansion	expans	expans	expand
history	histori	histori	hist
family	famili	famili	famy
poverty	poverti	poverti	poverty
rate	rate	rate	rat
entrepreneurship	entrepreneurship	entrepreneurship	entrepr
explosion	explos	explos	explod
more	more	more	mor
ever	ever	ever	ev
american	american	american	am
industry	industri	industri	industry
more	more	more	mor

第二部分 我自己的文本

第二段我随便写了点自己上周末的事，拿来试三个提取器：

2.1 原文（提取词干前）

Last weekend I finally cleaned my apartment, watered the dying plants on the balcony, and cooked something that actually tasted good for once. My roommate kept playing games and laughing loudly, so I gave up studying and joined him. We ordered too much food and

regretted it later. Honestly these small ordinary days are the ones I remember, not the busy ones full of running around and worrying.

2.2 三种提取器的输出

Porter 提取器：

last weekend i final clean my apart water the die plant on the balconi and cook someth that actual tast good for onc my roommat kept play game and laugh loudli so i gave up studi and join him we order too much food and regret it later honestli these small ordinari day are the one i rememb not the busi one full of run around and worri

Snowball 提取器：

last weekend i final clean my apart water the die plant on the balconi and cook someth that actual tast good for onc my roommat kept play gam and laugh loud so i gav up studi and join him we order too much food and regret it later honest these small ordinari day are the one i rememb not the busy one full of run around and worri

Lancaster 提取器：

last weekend i fin cle my apart wat the dying plant on the balcony and cook someth that act tast good for my room kept play gam and laugh loud so i gav up study and join him we ord too much food and regret it later honest thes smal ordin day ar the on i rememb not the busy on ful of run around and worry

2.3 三种提取器结果不一致的单词

共 68 个词，其中 36 个词被至少一种提取器改变，有 24 个词在三种算法下得到了不同的词干。部分示例：

原词	Porter	Snowball	Lancaster
finally	final	final	fin
cleaned	clean	clean	cle
watered	water	water	wat
dying	die	die	dying
balcony	balconi	balconi	balcony
actually	actual	actual	act
once	onc	onc	ont
roommate	roommat	roommat	room
games	game	game	gam
loudly	loudli	loud	loud
gave	gave	gave	gav
studying	studi	studi	study
ordered	order	order	ord
later	later	later	lat
honestly	honestli	honest	honest
these	these	these	thes
small	small	small	smal
ordinary	ordinari	ordinari	ordin

原词	Porter	Snowball	Lancaster
are	are	are	ar
ones	one	one	on
busy	busi	busi	busy
ones	one	one	on
full	full	full	ful
worrying	worri	worri	worry

第三部分 分析与讨论

提取词干前后有什么不一样？

对比下来最直接的感受就是：提取之后一堆词的尾巴被削掉了，-ing、-ed、-s、-ly、-tion 这些后缀基本都没了。比如里根那段里的 produce 和 produced 最后变成一个样子，rates 变成 rate。这其实挺有用的，做搜索或者词袋的时候，机器就不会把 produce 和 produced 当成两个不同的词，词表一下小了很多。

但缺点也很明显，就是提完之后很多词根本不是正常单词了。像 economy 变成 economi、history 变成 histori、family 变成 famili，看着挺别扭，人是读不下去的。所以词干提取是给机器用的，不是给人看的。

Porter、Snowball、Lancaster 三个有啥区别？

我一开始以为三个会差很多，结果发现 Porter 和 Snowball 几乎一模一样。查了下 Snowball 本来就是 Porter 的升级版（也叫 Porter2），所以大部分词两个结果都一样。偶尔能看出 Snowball 更聪明一点，比如我那段里 loudly 这个词，Porter 提成了 loudli，Snowball 直接给成 loud，明显后者更合理。

Lancaster 就完全是另一个画风了，砍得特别狠。像 roommate 被它砍成 room、actually 砍成 act、finally 砍成 fin，最离谱的是 once 被它弄成了 ont，这都快看不出原词是啥了。它有时候还砍得没规律，比如 dying 它反而没动，balcony 也没动，但一般的词又砍过头。课程里说它处理短词容易“造成混淆”，我算是亲眼看到了。

我觉得哪个更好？（这题没标准答案）

跑完这几组之后，我个人会选 Snowball。它跟 Porter 效果差不多，但个别地方处理得更干净，而且现在很多地方默认都用它。Porter 也不差，当个基础方案完全够用。Lancaster 我是真不太敢用，虽然它砍得多、词表压得最小，但像 room、act、ont 这种结果，意思都丢了，还容易把本来不一样的词混到一起。除非我就是想把词表压到最小、不在乎准不准，不然一般用不上它。当然作业也说了这题没对错，这就是我自己试下来的想法。